



OCEAN UNIVERSITY OF CHINA

EXPERIMENT 2

Personal Lab Report

Simulation Platform Setup
and Motion Planning

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Group:	5
Responsibility:	Simulation Platform Setup and Motion Planning

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Robotics Experiment

I Objectives

This experiment takes the mechArm270 6-degree-of-freedom robotic arm as the research object. We carried out robotic arm simulation and physical tests focusing on fixed-point pick-and-place tasks. The overall task of the group is to control the robotic arm to start from its home pose, grasp the predefined target object, transport it to the designated position, release the object, and finally return to the initial pose.

As Group Member 1, I was mainly responsible for the early-stage construction of the simulation platform, configuration of the robotic arm model, and verification of motion planning functions. My work laid the foundation for the subsequent development of fixed-point automatic grasping code and physical experiments.

Specific work:

- Install, configure and verify the availability of the Ubuntu 22.04 and ROS 2 Humble experimental environment.
- Build a robotic arm simulation environment based on Gazebo Fortress.
- Import and validate the mechArm270 robotic arm model.
- Test the motion performance of each joint and gripper of the robotic arm within the simulation environment.
- Configure relevant parameters for robotic arm motion planning based on MoveIt 2.
- Set the home pose of the robotic arm and test the normal operation of motion planning functions.
- Establish motion control communication between MoveIt 2 and Gazebo.
- Troubleshoot abnormal motion issues of the model, gripper and target object during simulation.
- Build a stable simulation platform for Member 2 to conduct subsequent fixed-point automatic grasping and repeated tests.

Through this experiment, I can further understand the composition of the robotic arm simulation system under ROS 2, master the working principle of MoveIt 2 motion planning, and become familiar with the complete experimental workflow from robot model loading to actual execution of planned trajectories.

II Individual Responsibilities

This experiment was completed by a four-person group, consisting of two main stages: simulation test and physical machine test. Member 1 and Member 2 were mainly responsible for the construction and testing of the simulation environment, while Member 3 and Member 4 were in charge of debugging and experiments on the real robotic arm.

As Member 1 of the group, I was primarily responsible for the basic construction of the simulation system. My work included configuring the ROS 2 Humble operating environment, building the Gazebo Fortress simulation platform, and importing the mechArm270 robotic arm model.

I also verified the model status and joint movement in RViz, configured relevant parameters for MoveIt 2, set the home pose of the robotic arm, and completed the verification of motion planning functions.

In addition, I verified the control connection between MoveIt 2 and Gazebo and analyzed abnormal motion problems during simulation.

After I completed the construction of the simulation platform and the verification of basic motion planning, Member 2 further configured key task parameters, including grasping point A, placing point B and the safe working height.

Member 2 then implemented the complete automatic grasping, transporting and placing process and conducted repeated simulation tests.

My work provided a stable, controllable and programmable simulation environment for the subsequent automatic grasping task.

III Experimental Environment and Main Software

Table 1: Experimental Environment and Main Software

Item	Experimental Configuration
Operating System	Ubuntu 22.04
ROS Version	ROS 2 Humble
Simulation Platform	Gazebo Fortress / Ignition
Visualization Tool	RViz2
Motion Planning	MoveIt 2
Control Framework	ros2_control
Robotic Arm	mechArm270
End-effector	Adaptive Gripper
Experiment Task	Fixed-point Pick-and-Place

ROS 2 handles information exchange among robot nodes, topics, controllers and other modules. Gazebo performs physical simulation of the robotic arm, workbench and target objects.

RViz is used to inspect the robot model, TF transforms and motion status. MoveIt 2 is mainly applied to the kinematic configuration and trajectory planning of the robotic arm.

IV ROS 2 and Gazebo Simulation Environment Setup

4.1 ROS 2 Humble Environment Verification

The experiment starts with verifying the running environment of ROS 2 Humble.

After sourcing the ROS 2 Humble environment in the Ubuntu terminal, the ROS_DISTRO environment variable and the path of the ros2 executable were checked.

The result showed that ROS_DISTRO was set to humble and the ros2 executable was available under the ROS 2 Humble installation path, confirming that the ROS 2 environment was successfully configured.

4.2 Gazebo Fortress Platform Verification

After confirming that the ROS 2 environment worked properly, Gazebo Fortress was launched.

Gazebo successfully rendered the 3D simulation world and displayed the entities and physical environment in the scene, indicating that the simulation platform could operate normally.

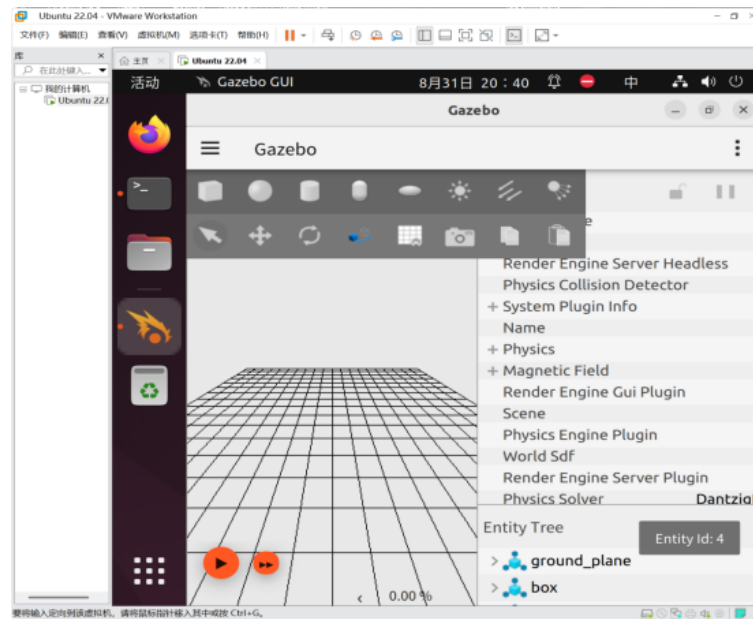


Figure 1: Gazebo Fortress simulation environment

This step established the basis for subsequent integration of the robotic arm model, controllers and MoveIt 2.

V mechArm270 Robotic Arm Model Configuration

5.1 RViz Model Loading

After finishing the basic configuration of ROS 2 and Gazebo, the mechArm270 robotic arm model was imported into the ROS 2 environment.

RViz was used to inspect the model.

The mechArm270 robotic arm could be rendered and displayed correctly, indicating that the robot description file was loaded properly and that the basic connections between the links were valid.

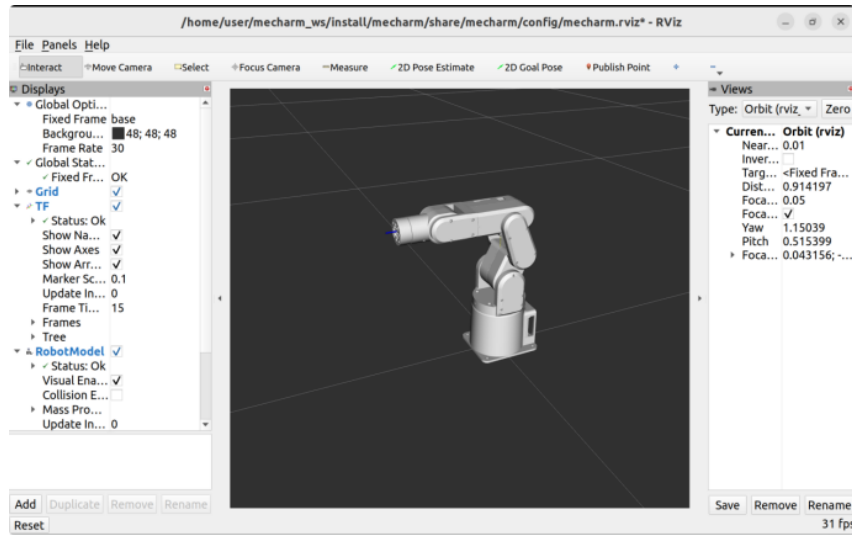


Figure 2: mechArm270 robotic arm model displayed in RViz

5.2 Joint Motion Verification

Successful display of the model alone cannot prove that all joint definitions are correct. Therefore, further joint-motion verification was performed.

The joint values were modified and the corresponding changes in robot pose were observed in RViz.

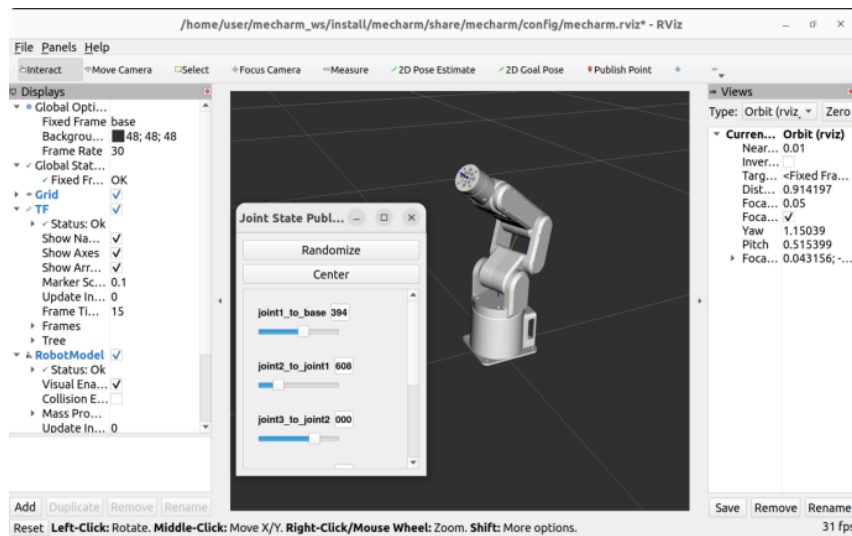


Figure 3: Joint motion verification using Joint State Publisher

The robotic arm pose changed according to the joint parameters, confirming that the basic joint definitions operated normally.

VI MoveIt 2 Motion Planning Configuration

6.1 Configuration via MoveIt Setup Assistant

After verifying the robotic arm model and basic joint motion, MoveIt Setup Assistant was used to configure motion planning for the mechArm270 robotic arm.

The configuration included robot planning groups, joint information, predefined poses and kinematic parameters.

6.2 Home Pose Configuration

To ensure a uniform starting state for each experiment, a home pose was defined in MoveIt 2.

The home pose serves as the reference position at the beginning and end of the task.

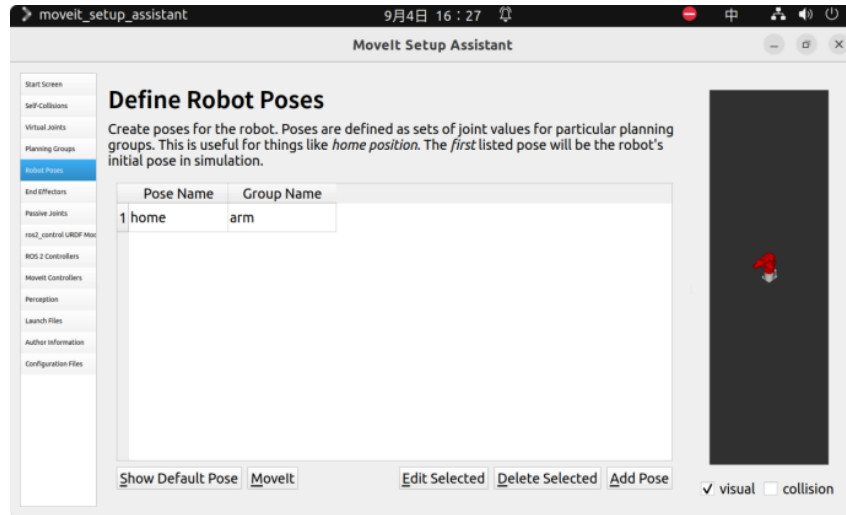


Figure 4: Home pose configuration in MoveIt Setup Assistant

6.3 MoveIt Configuration Package Generation

After completing the required settings, MoveIt Setup Assistant was used to generate the motion-planning configuration package.

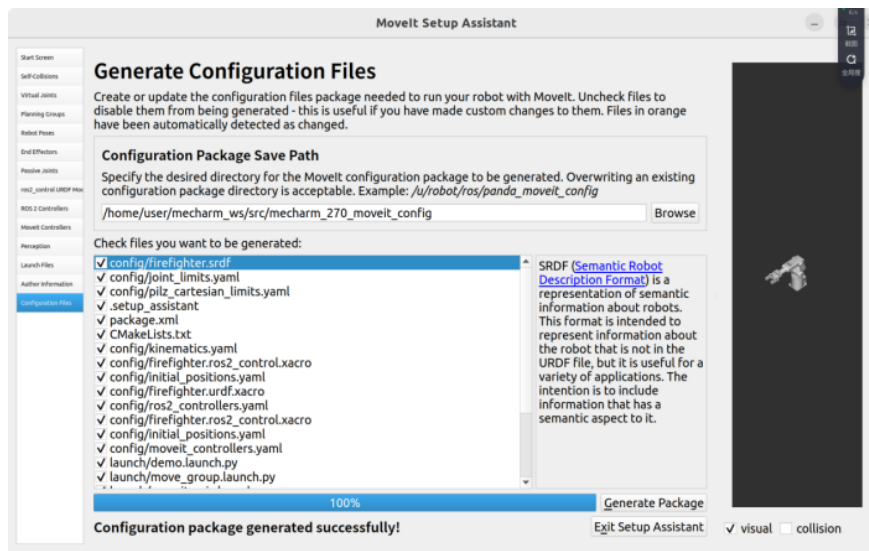


Figure 5: Generation of the MoveIt 2 configuration package

This package contains the main configuration files required for subsequent motion planning and control.

VII MoveIt 2 Motion Planning Verification

After completing the configuration, the MoveIt 2 MotionPlanning interface was launched in RViz.

The robotic arm planning group was selected and a target pose was defined. MoveIt 2 successfully generated a feasible planned trajectory.

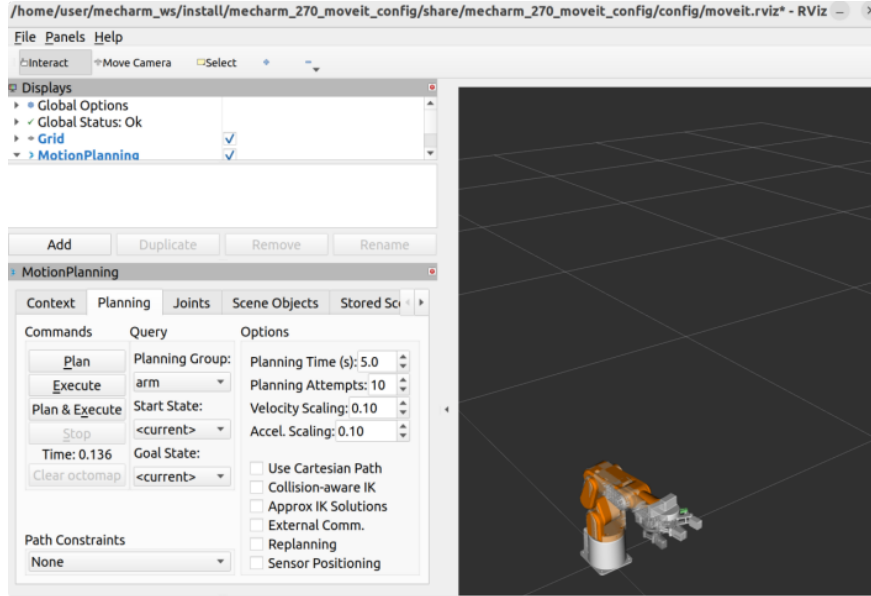


Figure 6: MoveIt 2 motion planning interface in RViz

Execution of the planned trajectory changed the robot pose according to the target, confirming that the motion planning function operated normally.

VIII Gazebo Work Scene Establishment

After completing the independent motion planning verification, a Gazebo work scene corresponding to the final grasping task was built.

The scene included the mechArm270 robotic arm, gripper, workbench, target object and ground.

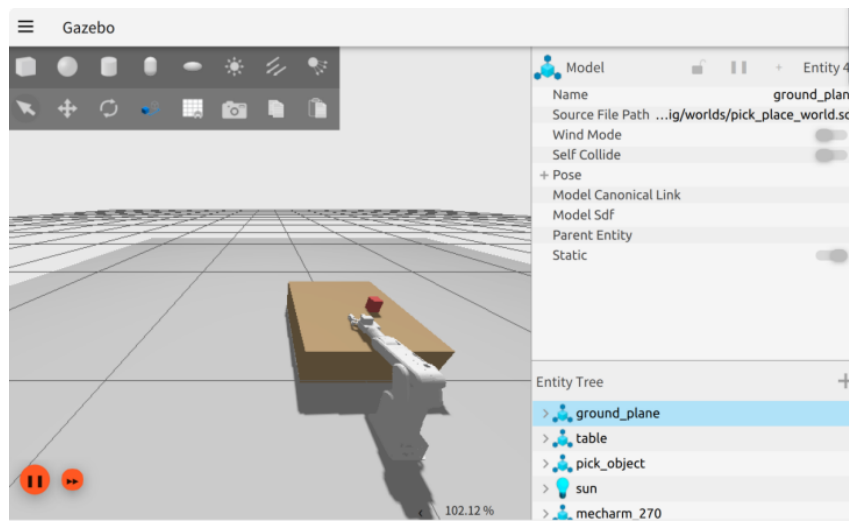


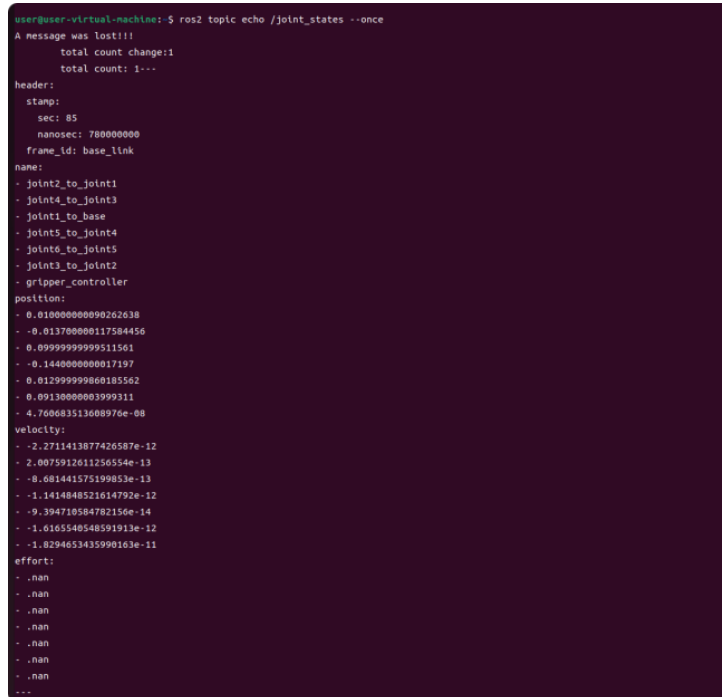
Figure 7: Gazebo work scene for the fixed-point grasping task

The addition of the workbench and target object transformed the simulation from simple robot motion verification into a practical fixed-point grasping environment.

IX ROS 2 Joint State Feedback Verification

After the robotic arm could execute movements, ROS 2 joint state feedback was further verified.

Joint information was read through the `/joint_states` topic.



```

user@user-virtual-machine:~$ ros2 topic echo /joint_states --once
A message was lost!!!
    total count change:1
    total count: 1---
header:
  stamp:
    sec: 85
    nanosec: 780000000
  frame_id: base_link
name:
- joint2_to_joint1
- joint4_to_joint3
- joint1_to_base
- joint5_to_joint4
- joint6_to_joint5
- joint3_to_joint2
- gripper_controller
position:
- 0.01000000000090262638
- -0.013700000117584456
- 0.099999999999511561
- -0.14400000000017197
- 0.0129999999800185562
- 0.091300000003999311
- 4.760683513608976e-08
velocity:
- -2.2711413877426587e-12
- 2.0075912611256554e-13
- -8.681441575199853e-13
- -1.1414848521614792e-12
- -9.394710584782156e-14
- -1.0165540548591913e-12
- -1.8294653435990163e-11
effort:
- .nan
- .nan
- .nan
- .nan
- .nan
- .nan
- .nan
---
```

Figure 8: ROS 2 joint state feedback from the `/joint_states` topic

The terminal returned data including joint names and corresponding positions, showing that a normal feedback channel had been established between the simulated robotic arm and ROS 2.

X Gripper Motion Verification

Besides movement of the robotic arm itself, the fixed-point grasping task requires the end gripper to grip and release the target object.

Corresponding control commands were sent in the simulation environment. The gripper responded with clear closing motion.

This demonstrated that the gripper model could respond to control commands and provided the basis for the later complete automatic grasping workflow.

XI MoveIt 2 and Gazebo Joint Control Verification

11.1 Purpose of Joint Control

MoveIt generates motion plans, while Gazebo performs physical simulation.

For the simulated grasping task, the planned trajectory generated by MoveIt must be received by the ROS 2 controller and executed by the robotic arm model in Gazebo.

11.2 Execution of Planned Trajectory

After launching MoveIt and Gazebo, a target trajectory was generated and sent to the robotic arm controller.

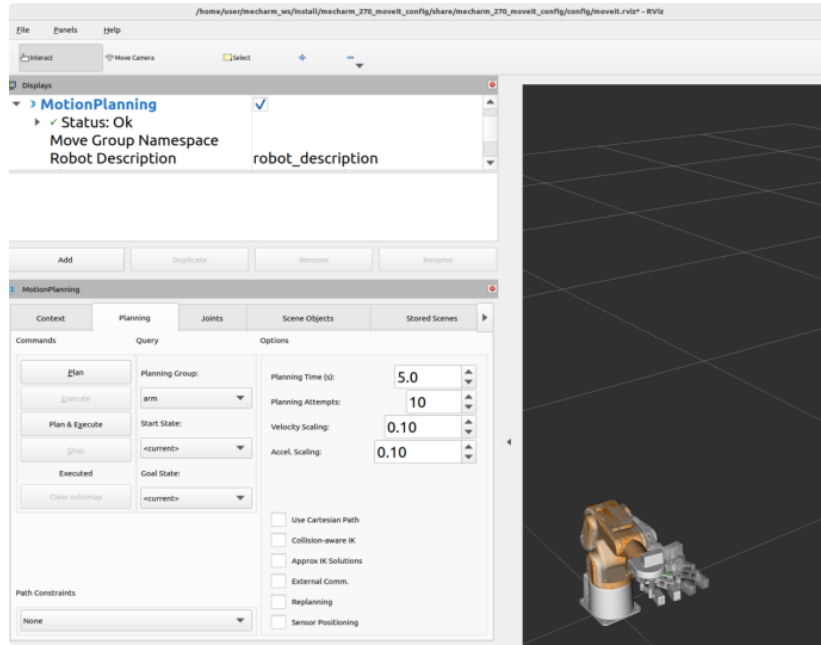


Figure 9: MoveIt 2 planning interface used for trajectory execution

The planned motion could be passed through the control pipeline, verifying the basic connection between motion planning and simulated robot execution.

XII Problems Encountered and Troubleshooting Process

12.1 Model Display and Control Verification

At the early stage of the experiment, the robotic arm model could already be displayed successfully in RViz.

However, successful visualization did not mean that the robot could execute control commands correctly. Therefore, joint motion, MoveIt planning, ROS 2 joint-state feedback and Gazebo trajectory execution were verified separately.

This layer-by-layer verification method made it easier to locate problems in individual modules.

12.2 Discrepancy Between Gripper Closing and Object Grasping Performance

Although the gripper could perform a closing action in simulation, actual grasping performance was also affected by gripper position, target-object placement and collision geometry.

In several tests, the gripper approached and closed around the target but the object did not move as

expected.

The gripper motion, object position, end-effector descending height, collision geometry and planned trajectory were therefore checked individually.

12.3 Abnormal Motion of Target Object

During simulation debugging, abnormal target-object poses and unsatisfactory grasping positions were observed.

This showed that successful motion planning alone could not guarantee stable grasping.

The position of the end-effector, collision range of the gripper and target-object position had to be coordinated.

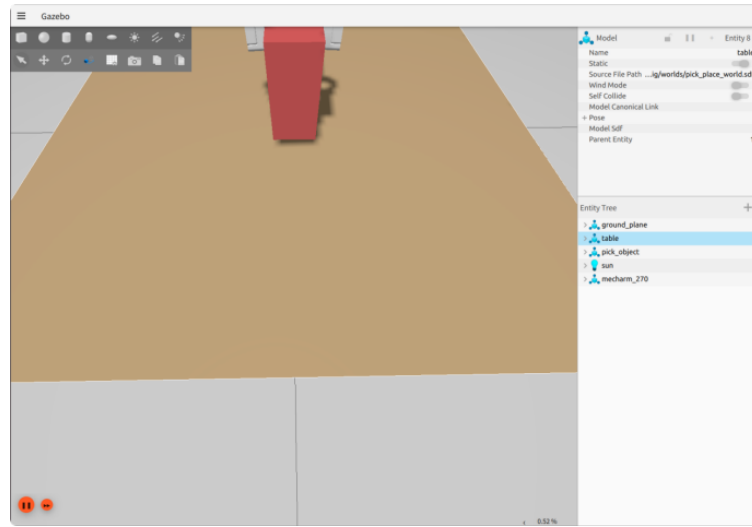


Figure 10: Target object position during grasping and debugging

12.4 Lifting the Grasped Object off the Table

After further adjustments, the target object could be lifted away from the table by the robotic gripper.

This showed that the robotic arm model, gripper control and target object could interact to produce a grasping effect.

It also provided a basis for Member 2 to further optimize point A, point B and the complete transportation workflow.

XIII Personal Experimental Results

Through the above experiments and debugging, I completed ROS 2 Humble environment verification, construction of the Gazebo Fortress simulation platform, mechArm270 model loading and joint inspection, MoveIt 2 configuration and home pose setup, motion planning verification, establishment of the grasping scene in Gazebo, ROS 2 joint state feedback verification, gripper motion testing, and control verification between MoveIt 2 and Gazebo.

Finally, a simulation environment capable of supporting subsequent development of the fixed-point automatic grasping task was established.

It provided the basis for Member 2 to realize grasping at point A, placing at point B, repeated grasping verification and anomaly tests.

XIV Team Collaboration and Work Handover

This experiment was completed collaboratively through task handover among different members.

My work was located at the front end of the complete experimental pipeline.

I first configured the robotic arm model and Gazebo simulation platform and verified the motion-planning and control pipeline using MoveIt 2.

After the basic environment operated stably, the simulation platform was handed over to Member 2 for further development of the fixed-point automatic grasping task.

Member 2 further configured grasping point A, placing point B and the safe height.

The complete workflow was:

Home → above point A → descend → grip → lift → above point B → descend → release → Home

Repeated execution and anomaly tests were then carried out.

After completion of the simulation stage, Member 3 and Member 4 were mainly responsible for experiments and verification on the physical robot.

Therefore, my main contribution was to build and verify the basic robotic simulation and motion-planning platform required for the subsequent stages of the project.

XV Personal Experience and Insights

Through this experiment, I gained a more intuitive and systematic understanding of the ROS 2 robotic arm simulation system compared with pure theoretical learning.

At the beginning of the experiment, I assumed that successful model display in RViz or Gazebo meant that the simulation environment had essentially been completed.

However, practical debugging showed that a complete robotic simulation system contains multiple coupled modules, including robot model parsing, joint-state feedback, motion planning, controller configuration, Gazebo physical simulation and ROS 2 communication.

Model visualization only verifies that the robot description can be loaded; it does not prove that the robotic arm can correctly execute a task.

During joint debugging of MoveIt 2 and Gazebo, I also developed a clearer understanding of the difference between motion planning and motion execution.

MoveIt 2 generates feasible planned trajectories according to the current and target poses of the robotic arm, while actual movement in Gazebo relies on the coordinated operation of ROS 2 controllers, interfaces and the physical simulation model.

The gripper experiments also showed me an important difference between robotics experiments and conventional programming.

The absence of a program error does not guarantee a successful physical result.

Even if the gripper closes correctly, grasping failure or object offset may still occur because of target placement, collision-model accuracy or the trajectory posture.

Therefore, robotic experiments require observation of both terminal information and the actual physical behavior shown in simulation.

This experiment also helped me develop a more systematic troubleshooting method.

Instead of modifying several parameters at the same time, it is more effective to verify the model, joint state, motion planning, control execution and grasping effect layer by layer.

By completing the construction and verification of the simulation platform, I became familiar with ROS 2 Humble, Gazebo Fortress, RViz and MoveIt 2 and developed a clearer understanding of the complete control pipeline from model loading to trajectory execution.

The experiment also strengthened my understanding of teamwork.

The simulation platform I established served as the basis for the subsequent automatic grasping task and physical robot experiments. Only through effective connection between the modules completed by different members could the complete fixed-point grasping task be successfully achieved.

XVI Conclusion

In this experiment, I was mainly responsible for construction of the ROS 2 and Gazebo simulation environment, configuration of the mechArm270 robotic arm model, MoveIt 2 motion-planning debugging, and control verification between MoveIt 2 and Gazebo.

My work included ROS 2 environment verification, Gazebo platform deployment, RViz model visualization, joint-motion testing, MoveIt 2 parameter configuration, home-pose setting, motion-planning verification, ROS 2 joint-state feedback, gripper control and simulated trajectory execution.

In addition, I analyzed abnormal object movement and unstable grasping behavior during simulation.

Finally, I successfully established and verified the basic simulation and motion-planning platform for the fixed-point grasping task.

This work provided a stable technical foundation for subsequent automatic grasping programming, point-position optimization and physical robot verification experiments.